

Data-driven collaborative optimization of three-dimensional highway alignments for life-cycle cost and mixed fuel–electric traffic energy

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Abstract

Highway alignment couples plan geometry, profile, terrain, structures, urban development and long-term vehicle operation. We develop a spatially explicit bi-objective framework combining a quasi-natural digital elevation model, OpenStreetMap transport, water and buildings, and 14,018 GNSS points. Fifty plan modes and 224 referenced grade genes are searched collaboratively; centering leaves at most 223 locally independent grade degrees of freedom. Life-cycle cost includes land, basic works, structures, earthwork and maintenance, while the second objective monetizes 30-year fuel and electricity use for a mixed fleet. Spatial intersections trigger bridges, a connected mountain zone triggers ecological tunnelling, and dense cores are excluded. Improved Jellyfish Search approximates a weighted-sum front; feasibility, budget and non-dominance gates precede entropy-based selection.

The 22.46-km Guangzhou North Ring case anchors endpoint elevations, activates density constraints and uses spectral width $W = 500$ m. The selected alignment reduces cost from 2.6406 to 2.4208 billion RMB (8.33%) and energy cost from 1.3946 to 1.3528 billion RMB (3.00%), with a 421-m minimum radius, zero hard penalty and no Tier-2 exposure. Tier-1 exposure falls from 2.010 to 1.591 km. The route is 12 m longer and uses more earthwork, but structure cost falls by 237 million RMB, exposing a physical trade-off rather than a length artefact. Earlier free-endpoint, unequal-budget experiments remain exploratory; current-model multi-seed, equal-NFE replication is required before design-grade use.

Keywords: highway alignment optimization, digital elevation model, OpenStreetMap, improved Jellyfish Search, life-cycle cost, mixed traffic energy, horizontal–vertical coordination

1. Introduction

Highway alignment is an early design decision with long-lived consequences. A small lateral shift changes the terrain crossed, land take, bridge and tunnel demand, earthwork balance, maintenance exposure and the grades experienced by vehicles. These effects are coupled: shortening a horizontal path may worsen the vertical profile; avoiding a mountain may enter a dense urban zone; lowering earthwork can require a bridge; and a profile that appears energy-efficient in one direction can become infeasible at its network tie-ins. The practical problem is therefore not a curve-fitting exercise but a constrained, spatially explicit, multi-objective design problem.

Classical formulations established the value of simultaneous horizontal and vertical optimization [1, 2]. Subsequent research introduced corridor models, mixed-integer formulations, environmental objectives and preference-aware multi-objective search [3, 4, 5, 6, 7]. Recent work has coupled alignment generation with BIM, GIS, deep reinforcement learning, sampling-based planning, and hybrid reinforcement-learning/metaheuristic search [8, 9, 10, 11, 12]. A comprehensive review by Song et al. [13] nevertheless identifies enduring gaps between algorithmic models and the heterogeneous spatial constraints encountered in engineering design.

Three gaps motivate this study. First, many models use a single centerline terrain profile or stylized land-cost grid. This prevents horizontal displacement from changing the terrain and can erase the main economic reason for joint optimization. Second, bridges, tunnels and urban avoidance are commonly imposed as fixed lengths or discontinuous penalties. Such simplifications allow an optimizer to exploit bookkeeping rules rather than discover an engineering alternative. Third, large nonlinear decision spaces remain difficult for population-based solvers. A solver may report a low objective while retaining localized curvature or grade-change violations that disappear in an averaged penalty.

This paper extends the mathematical base in Lin [14] while replacing its NSGA-II-centered and centerline-based implementation with a current, data-driven experiment. The main contributions are:

- (1) a spatially explicit digital corridor that fuses GNSS, quasi-natural DEM, OSM transport/water features and OSM-derived building-density tiers;

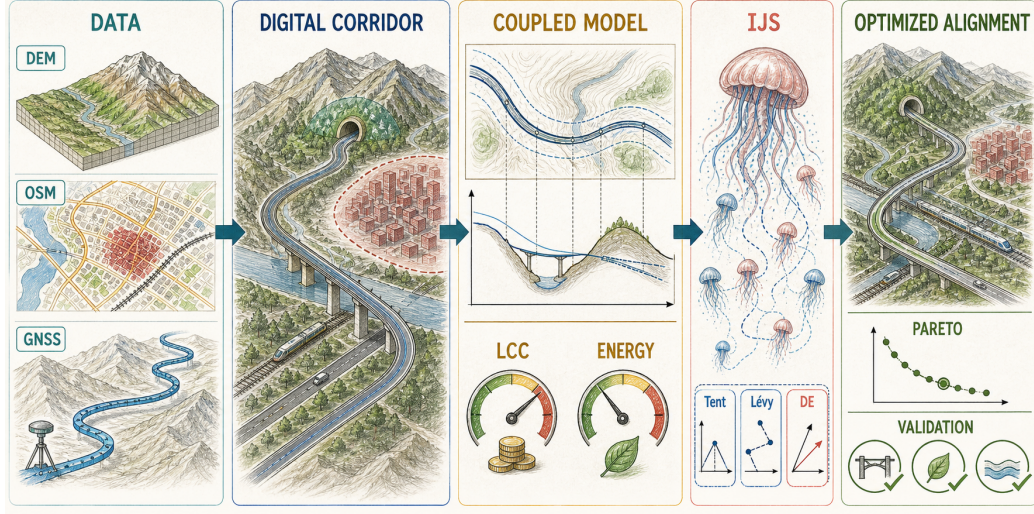


Figure 1: Data-driven workflow for collaborative three-dimensional highway alignment optimization. The hand-drawn scientific illustration highlights the information flow; the mathematical definitions are given in Sections 3 and 4.

- 35 (2) an engineering-oriented life-cycle model in which crossing bridges, ecological
36 tunnelling, earthwork-to-structure substitution and endpoint tie-ins
37 are endogenous to the three-dimensional geometry;
- 38 (3) a nominally 275-slot collaborative parameterization with 274 referenced
39 genes and at most 273 locally independent degrees of freedom, coupled
40 to a bi-objective cost–energy decision process;
- 41 (4) an IJS solver that combines corrected Tent initialization, domain-scaled
42 Lévy exploration and differential-evolution refinement; and
- 43 (5) an auditable evidence chain that separates the current engineering result
44 from exploratory experiments conducted during earlier model develop-
45 ment.

46 Figure 1 summarizes the proposed pipeline. The visual distinction be-
47 tween data, digital corridor, coupled model, optimizer and validated align-
48 ment is intentional: each block has an auditable counterpart in the experi-
49 ment repository.

2. Related research

2.1. Three-dimensional alignment optimization

Early simultaneous models demonstrated that horizontal and vertical decisions should not be optimized independently [1]. Evolutionary encoding subsequently made non-convex three-dimensional search practical [2]. GIS-based studies expanded the feasible region and allowed environmental layers to influence routing [3, 15]. Corridor-constrained horizontal search [4] and rigorous vertical-alignment formulations [7] improved mathematical tractability, while bi-objective road and railway models incorporated cost, hazard and ecological impacts [5, 16, 17].

Recent end-to-end models integrate BIM or intelligent path planners, while newer hybrid schemes combine policy optimization with population-based search [8, 11, 12]. These methods improve automation but do not eliminate the need for transparent engineering accounting. In particular, a spatial intersection must lead to a structure or clearance decision, terrain sampling must follow the candidate line, and existing network connections must be preserved. This paper therefore prioritizes traceable geometry-to-cost mappings over an opaque surrogate objective.

2.2. Life-cycle cost and traffic energy

The underlying life-cycle model follows the categories in Lin [14]: land acquisition, bridges/tunnels, road works, maintenance and earthwork. Vehicle operation is relevant because geometric consistency affects emissions and energy demand [18]. Electric-vehicle energy is especially sensitive to grade, mass, auxiliaries and regenerative behavior [19]. Rather than adding fuel and electricity in incompatible physical units, the present framework monetizes both over the analysis period. This does not claim that money is a complete environmental metric; it supplies a consistent decision scale while physical fuel and electricity components remain reportable.

2.3. Optimization algorithms and objective decision

NSGA-II remains a standard multi-objective benchmark [20], but direct evolutionary search becomes expensive as the number of profile variables increases. Jellyfish Search models ocean-current and active/passive motion [21]. The present IJS adds three complementary mechanisms: a Tent map for initial coverage, Lévy flights for non-local exploration and differential evolution (DE) for vector-wise refinement [22]. Section 5.3 reports historical

85 same-generation exploratory associations for these mechanisms; confirmation
 86 on the current formulation requires equal-NFE reruns.

87 Entropy weighting is based on the information concept of Shannon [23].
 88 Here it is applied only after feasibility and non-dominance screening. Thus a
 89 high weight cannot compensate for a geometric violation, and an infeasible
 90 solution cannot be selected merely because its cost is low.

91 3. Data-driven mathematical formulation

92 3.1. Study corridor and data fusion

93 The case is the Xunfengzhou–Huangcun portion of Guangzhou North
 94 Ring Expressway, a two-way six-lane urban expressway with a measured
 95 centerline length of 22.462 km. The input comprises 14,018 GNSS points
 96 with plan coordinates and elevations. These points define the existing scheme
 97 M-A, the fixed plan endpoints and the vertical tie-in elevations.

98 Terrain is sampled from AWS Terrain Tiles [24]. Sixty zoom-level-14
 99 tiles are decoded at approximately 8.8 m pixel spacing. Because the avail-
 100 able surface represents the built road, a ± 60 -m road mask is removed and
 101 interpolated from neighboring terrain to obtain a quasi-natural ground. Its
 102 correlation with the measured centerline elevation is 0.915 after a constant
 103 vertical-datum correction. This reconstruction is imperfect but applies the
 104 same terrain accounting to the existing and optimized alternatives.

105 OSM [25, 26] supplies roads, railways, watercourses and building foot-
 106 prints. After removing the North Ring Expressway and parallel facilities,
 107 the obstacle set contains 4,807 road features, 857 railway features and 338
 108 river/canal features. Building footprints are rasterized on a 25-m grid and
 109 smoothed with a 200-m kernel. Thresholds are calibrated against the max-
 110 imum density already traversed by M-A: Tier 2 is a hard exclusion, Tier 1
 111 is passable with a soft discouragement, and Tier 0 is unrestricted. Figure 2
 112 shows the aligned layers.

113 3.2. Collaborative decision variables

114 Let the measured plan line be $\mathbf{r}_0(s) = [x_0(s), y_0(s)]^\top$ with unit normal
 115 $\mathbf{n}(s)$, $s \in [0, L_0]$. A candidate plan is represented by a normal offset

$$\delta(s) = \sum_{k=1}^{50} a_k \sin\left(\frac{k\pi s}{L_0}\right), \quad |a_k| \leq \frac{W}{k^2}, \quad (1)$$

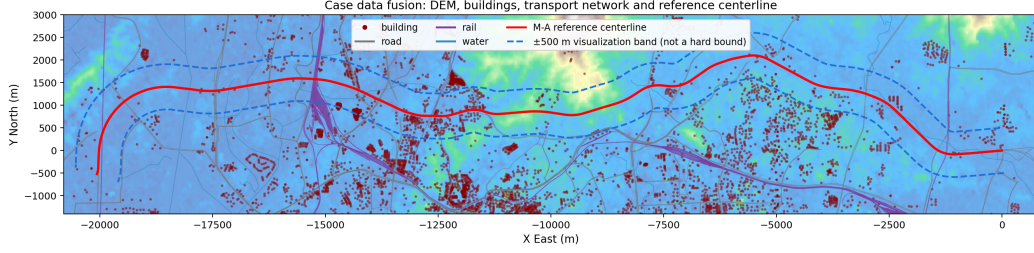


Figure 2: Case-study data layers in the local engineering coordinate system: DEM, OSM buildings, roads, railways, watercourses, measured centerline and a displayed ± 500 -m context band. In the optimizer W is the first sine-mode amplitude, not a pointwise hard offset bound.

116 where $W = 500$ m in the main experiment and $\mathbf{r}(s) = \mathbf{r}_0(s) + \delta(s)\mathbf{n}(s)$.
 117 The sine basis fixes both endpoints and attenuates high-frequency curvature.
 118 It implies the conservative spectral envelope $|\delta| \leq W \sum_{k=1}^{50} k^{-2} \approx 1.625W$;
 119 hence W is not a pointwise corridor boundary. The selected M-C has a
 120 maximum sampled offset of approximately 184 m. Offset control points are
 121 reconstructed by a cubic spline and evaluated on a 10-m grid. For recon-
 122 structed coordinates $(x(s), y(s))$,

$$\kappa(s) = \frac{|x'(s)y''(s) - y'(s)x''(s)|}{[x'(s)^2 + y'(s)^2]^{3/2}}, \quad R(s) = \kappa(s)^{-1}. \quad (2)$$

The vertical profile uses $M = 225$ stations (approximately 100 m apart). Normalized variables generate segment grades, but the start and end design elevations z_0 and z_L are tied to the measured road. For raw grade perturbations $h_i \in [-1, 1]$, segment lengths Δs_i and maximum grade $g_{\max} = 0.04$, define

$$\bar{h} = \frac{\sum_i h_i \Delta s_i}{\sum_i \Delta s_i}, \quad g_{\text{req}} = \frac{z_L - z_0}{\sum_i \Delta s_i}, \quad (3)$$

$$g_i = g_{\text{req}} + \lambda g_{\max}(h_i - \bar{h}), \quad (4)$$

123 where $\lambda \in [0, 1]$ is the largest scaling that retains $|g_i| \leq g_{\max}$. Consequently
 124 $\sum_i g_i \Delta s_i = z_L - z_0$ exactly. The implementation that generated the au-
 125 thoritative case alignment retains a legacy start-elevation gene in its 225-slot
 126 profile block, but ignores it when z_0 is tied. Its stored vector therefore has 275
 127 slots and 274 referenced genes (50 modes and 224 grades). Because adding
 128 a common constant to all 224 raw grades is removed by centering, it has at

most 273 locally independent degrees of freedom (50+223); the roughly 150 reconstruction controls are not optimization variables.

The confirmatory experiments remove both redundant directions without changing the decoder. For raw grade genes $r \in [0, 1]^{224}$, define the quotient coordinates

$$q_i = r_i - r_{224} + \frac{1}{2}, \quad i = 1, \dots, 223, \quad q_i \in [-\frac{1}{2}, \frac{3}{2}], \quad (5)$$

and reconstruct $r' = (q_1, \dots, q_{223}, 1/2)$. Since r' differs from r only by a common translation, subtracting the mileage-weighted mean gives exactly the same centred grade vector. The ignored start slot is fixed at 1/2. Thus the confirmation searches 50 plan modes plus 223 grade contrasts, i.e., 273 non-redundant coordinates, while retaining the same endpoint anchoring, grade scaling, DEM/OSM queries and engineering objectives.

3.3. Life-cycle cost

The first objective is

$$\min C = C_R + C_B + C_S + C_Q + C_{TU}, \quad (6)$$

where C_R is land acquisition, C_B bridge/tunnel cost, C_S basic subgrade and pavement cost, C_Q discounted maintenance and C_{TU} earthwork/haulage. With road width B , length L , one mu = 666.67 m², bridge/tunnel lengths L_B, L_T and invariant ramp cost C_{ramp} , the implemented accounting is

$$C_R = c_{land} \frac{LB}{666.67} + 5000 \frac{L}{1000}, \quad (7)$$

$$C_B = c_B L_B + C_{ramp} + c_T L_T, \quad C_S = (c_{sg} + c_{pv})L, \quad (8)$$

$$C_Q = \sum_{y=1}^T \frac{1}{(1+r)^y} \sum_i \Delta s_i [\gamma + 365 Q_y \tau + c_{soil} \Phi_i]. \quad (9)$$

Here Φ_i is the weighted cut/fill slope-exposure term in the implementation; it is zero on structure-exempt segments. Unit rates follow the source thesis and the 2025 Guangzhou governmental estimation guideline [27]. The analysis period is 30 years with a 5% discount rate.

At station j , let $h_j = z_j - z_j^g$ be the design-to-ground height, B the formation width and m the side-slope ratio. The approximate cross-sectional earthwork area is

$$A_j = B|h_j| + mh_j^2. \quad (10)$$

Volumes are integrated by the average-end-area method. Define the station cost rate

$$V^\pm = \sum_j \frac{A_j^\pm + A_{j+1}^\pm}{2} \Delta s_j, \quad (11)$$

$$c_j = \begin{cases} c_{str}, & |h_j| > 30 \text{ m or } c_{ew}A_j > c_{str}, \\ c_{ew}A_j, & \text{otherwise,} \end{cases} \quad (12)$$

$$C_{TU} = E_L + \sum_j \frac{c_j + c_{j+1}}{2} \Delta s_j, \quad (13)$$

where the sign of h_j selects the fill/bridge or cut/tunnel pair of rates. Structure-exempt stations have $c_j = 0$. If fill/cut cost exceeds the corresponding bridge/tunnel cost, or if $|h_j| > 30$ m, structural substitution is triggered. The selected structure cost replaces rather than supplements earthwork in that interval, preventing double counting and “free floating road” solutions.

For each candidate plan, intersections with OSM road classes primary and above, railways and watercourses trigger bridges. If θ_q is the crossing angle and b_q the nominal obstacle width, the required crossing interval is

$$\ell_q = \frac{b_q}{\max(\sin \theta_q, \sin 15^\circ)} + 2E_{\text{ext}}, \quad (14)$$

where $E_{\text{ext}} = 75$ m is the approach extension. Gaps under 100 m are merged. Ramp-function cost is calibrated from the seven existing interchanges and kept invariant, whereas mainline bridge length is geometric. A connected terrain region above 70 m, closed morphologically over 500 m, defines the Baiyun ecological mandatory-tunnel zone. Its area (19.2 km²) and M-A crossing length (1.28 km) agree with independent project information (about 21 km² and 1.35 km, respectively).

3.4. Mixed fuel–electric traffic energy

The second objective monetizes lifetime operation of a mixed fleet:

$$\min E = \sum_{y=1}^{30} \frac{365 Q_y}{(1+r)^y} [(1 - p_{EV})c_f e_f(\mathbf{x}) + p_{EV}c_e e_e(\mathbf{x})], \quad (15)$$

Table 1: Principal current-model parameters used in the case study.

Group	Parameter	Value
Geometry	speed / width / R_{min} / g_{max}	100 km/h / 25.5 m / 400 m / 4%
Life cycle	T / discount rate	30 yr / 5%
Land/basic works	c_{land} / c_{sg} / c_{pw}	58,641 RMB/mu / 200 / 30,000 RMB/m
Structures	c_B / c_T / E_{ext}	156 / 270 million RMB/km / 75 m
Earthwork	cut / fill / side slope	30 / 25 RMB/m ³ / 1:1.5
Traffic	AADT / EV share	30,000 veh/day / 0.30
Energy	fuel / electricity price	8 RMB/L / 0.8 RMB/kWh
Search	population / iterations / weights	200 / 1,000 / 21

167 where Q_y is AADT, $p_{EV} = 0.30$, c_f and c_e are fuel and electricity prices, and
168 e_f, e_e are per-vehicle route consumption. The baseline uses AADT=30,000
169 vehicles/day, gasoline/electricity prices of 8 RMB/L and 0.8 RMB/kWh, and
170 the same discount rate as cost.

For a fuel vehicle, the traction load includes aerodynamic, rolling, grade and residual lateral forces. With speed v , mass m_v , aerodynamic coefficient C_d , frontal area A_f , grade angle α and curve radius R , these are

$$F_{air} = \frac{1}{2}\rho C_d A_f v^2, \quad F_r = C_r m_v g \cos \alpha, \quad (16)$$

$$F_g = m_v g \sin \alpha, \quad F_a = \max \left(0, \frac{m_v v^2 / R - m_v g S_p}{N_v C_s} \right). \quad (17)$$

For fuel vehicles the implementation converts external load to horsepower and fuel volume as

$$P_{ex,i} = \max \left(0, \frac{(F_{air} + F_r + F_g + F_a)v}{736\eta_f} \right), \quad (18)$$

$$e_f(\mathbf{x}) = \sum_i \phi(HP_{in} + P_{ex,i}) \frac{\Delta s_i}{v} / 1000, \quad (19)$$

171 where e_f is L/vehicle-trip, $\phi = 0.03$ ml/(s hp), $HP_{in} = 27$ hp and $\eta_f =$
172 0.85 . The printed 10^{-3} multiplier in the source expression for F_a is omitted:
173 retaining it makes lateral resistance roughly five orders too small and removes
174 the plan-curvature energy channel. This dimensional correction is explicitly
175 disclosed rather than silently absorbed into calibration.

176 Electric consumption uses signed segment work $W_i = (F_{air} + F_r + F_g +$

177 $F_a)\Delta s_i$:

$$e_e(\mathbf{x}) = \left[\frac{1}{3.6 \times 10^6} \sum_i \begin{cases} W_i/\eta_e, & W_i \geq 0, \\ W_i\eta_e, & W_i < 0, \end{cases} \right]_+ + 0.15L_{km}, \quad (20)$$

178 where $[x]_+ = \max(x, 0)$ and $\eta_e = 0.90 \times 0.95$. Fuel and electric use are re-
 179 tained separately before monetary aggregation. The current main experiment
 180 evaluates a representative travel direction; fixing both endpoint elevations re-
 181 moves the former net-downhill bookkeeping exploit. A bidirectional mean is
 182 identified as a required extension in Section 6.4.

183 3.5. Constraints and decision rule

184 The hard engineering set includes fixed plan/profile endpoints, $R(s) \geq$
 185 400 m for 100 km/h design speed, $|g_i| \leq 4\%$, $|g_{i+1} - g_i| \leq 3 \times 10^{-4}\Delta s_{ctrl}$,
 186 crossing skew angle $\theta \geq 15^\circ$, and zero Tier-2 penetration. Values follow JTG
 187 D20-2017 [28] where directly applicable. W controls sine-mode amplitudes
 188 but is not a pointwise corridor constraint; the selected current line remains
 189 inside the displayed ± 500 -m context band. Local violations use the sum
 190 of mean and maximum relative violation, so a short severe defect cannot
 191 disappear in a long route average.

192 For any relative-violation vector $\mathbf{v}$, define $H(\mathbf{v}) = \text{mean}(\mathbf{v}) + \max(\mathbf{v})$.
 193 The implemented penalty is

$$P = \rho_t \{30H(\mathbf{v}_R) + 10H(\mathbf{v}_{\Delta g}) + 10H(\mathbf{v}_{skew}) + 10H(\mathbf{v}_{D2})\}, \quad (21)$$

194 where $v_R = [400 - R]_+/400$, $v_{\Delta g}$ and v_{skew} are normalized excesses, and
 195 $v_{D2} = \min(d_{D2}/100, 5)$. The schedule is $\rho_t = 0.3$ in the first half and 3.0
 196 thereafter. Tier 1 is deliberately excluded from P and appears only as the
 197 soft term in Eq. (22).

198 For objectives C and E , a common baseline population supplies reference
 199 scales C_{ref} and E_{ref} . Entropy weights w_C, w_E produce

$$F(\mathbf{x}) = w_C \frac{C(\mathbf{x})}{C_{ref}} + w_E \frac{E(\mathbf{x})}{E_{ref}} + P(\mathbf{x}) + 0.22 \frac{L_{T1}(\mathbf{x})}{L(\mathbf{x})}. \quad (22)$$

200 For an $n \times 2$ objective matrix, min-max normalized entries z_{ij} give $p_{ij} =$
 201 $z_{ij} / \sum_i z_{ij}$, entropy $e_j = -(\ln n)^{-1} \sum_i p_{ij} \ln(p_{ij} + 10^{-12})$, and $w_j = (1 -$
 202 $e_j) / \sum_k (1 - e_k)$. Weights computed from a common initial population define

the search scalarization. They are recomputed on the surviving front for the final compromise; these are two distinct uses of entropy weighting.

IJS solves 21 independent weighted-sum problems, so the resulting set is a weighted-sum Pareto approximation and can miss unsupported points on a non-convex front. The final decision is selected by four gates: hard feasibility, project-screening budget $C \leq 1.1C_{M-A}$, non-dominance, and entropy-weighted compromise. The 10% budget margin is an explicit planning assumption rather than a statutory threshold; its decision sensitivity remains to be tested.

4. Improved Jellyfish Search

4.1. Mechanisms

The original JS alternates an ocean-current update toward the best solution with active/passive movement within the population [21]. The IJS implementation adds:

- **Independent Tent initialization.** Each individual uses an independent chaotic chain with $\mu = 1.99$. The exact value $\mu = 2$ is avoided because finite-precision binary arithmetic collapses to zero; a single shared chain is avoided because it creates correlated individuals.
- **Domain-scaled Lévy flight.** The heavy-tailed perturbation is scaled by $0.01(\mathbf{u} - \mathbf{l})$. The former unscaled implementation had only a 0.48% acceptance rate because a unit step is not meaningful across heterogeneous variable ranges.
- **DE refinement.** Mutation and crossover with $CR = 0.5$ form a trial vector from population differences, followed by elitist acceptance [22].

For reproducibility, the time control is $A_r = (1 - t/T)(2u - 1)$. If $|A_r| \geq 0.5$, the main trial is $\mathbf{x}_i + \mathbf{u} \odot (\mathbf{x}_{best} - 3u\bar{\mathbf{x}})$; otherwise the code applies active motion toward a fitter random peer or passive motion $\mathbf{x}_i + 0.1\mathbf{u} \odot (\mathbf{u}_b - \mathbf{l}_b)$. The Lévy trial is

$$\mathbf{x}_i^L = \mathbf{x}_i + \mathbf{u} \odot [0.01(\mathbf{u}_b - \mathbf{l}_b) \odot \mathbf{L}_{1.5}], \quad (23)$$

and the DE trial uses $\mathbf{m}_i = \mathbf{x}_i + u(\mathbf{x}_{r1} - \mathbf{x}_{r2})$, binomial crossover with $CR = 0.5$, and greedy replacement. Out-of-bound coordinates are reflected

233 across the opposite bound and then clipped. Every main, Lévy and DE trial
234 is evaluated before elitist acceptance.

235 Each IJS generation therefore has about three population-equivalent eval-
236 uation phases; this is reported rather than comparing equal generation counts
237 as equal effort. The authoritative single case alignment used a two-phase
238 penalty schedule, with 0.3 of the nominal penalty in the first half and 3.0 in
239 the second half, to explore near feasibility boundaries before recovery. The
240 equal-NFE confirmation instead uses the static hard target defined below for
241 every algorithm. Figure 3 gives the original case-solution process.

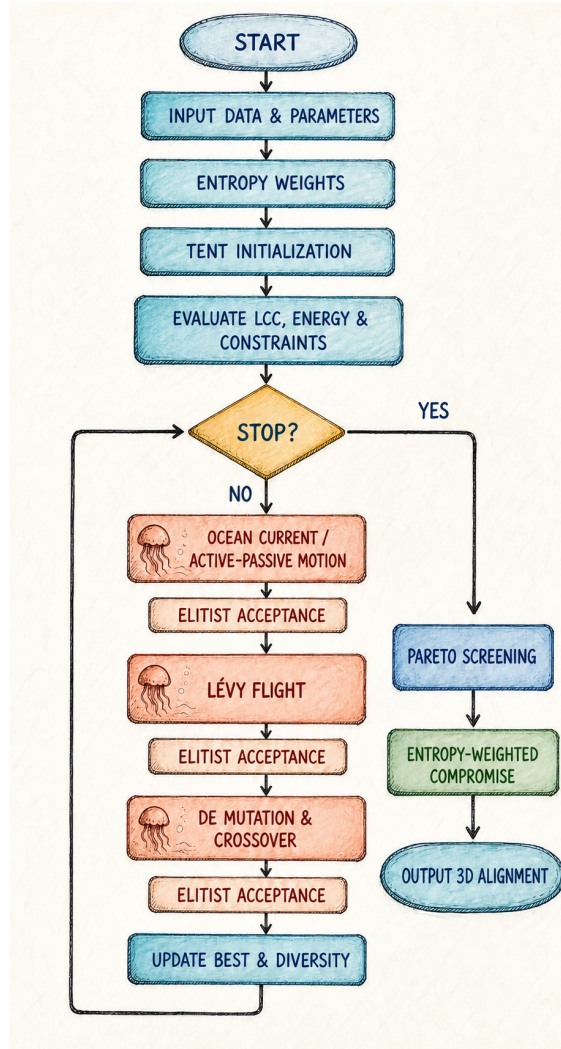


Figure 3: IJS solution and decision flow. Each operator is followed by elitist acceptance; the stopping branch proceeds to Pareto screening and entropy-weighted compromise selection, not geometric knee selection.

242 4.2. Experimental protocol

243 The scheme experiment uses population 200, 1,000 IJS iterations and 21
 244 Pareto weights (a later visual sweep uses 40). Objectives are evaluated at
 245 approximately 10-m spacing. The baseline, cost-only and joint alternatives
 246 use the same data and reference scales.

247 The historical ablation fixes the measured plan and uses a nominal 225-
 248 variable free-elevation profile model. Five variants (JS, +Tent, +Lévy, +DE
 249 and complete IJS) have 30 runs, population 200 and 500 iterations. The
 250 historical benchmark compares IJS, JS, NSGA-II, real-coded GA, PSO and
 251 GWO at six profile spacings (95–500 stored variables), with 10 seeds per pair.
 252 Reported p values come from independent-sample Mann–Whitney/rank-sum
 253 tests, not paired signed-rank tests. These experiments predate endpoint an-
 254 choring and density tiers. Equal generations also imply unequal NFE: approx-
 255 imately 100,200 for JS, 200,200 for one-added-operator variants and 300,400
 256 for full IJS in the 500-generation ablation. They are therefore retained only
 257 as transparent development evidence and cannot establish equal-budget su-
 258 periority of IJS for the current model. The same boundary applies to the
 259 237 historical re-optimization sensitivity points.

260 The confirmatory experiment freezes the current endpoint-anchored,
 261 DEM/OSM and density-enabled model, the entropy weights $(w_C, w_E) =$
 262 $(0.50838, 0.49162)$ and reference scales from the authoritative case result.
 263 It uses the non-redundant 273-coordinate representation, population 200
 264 and exactly 600,000 objective calls per run. Every Tent, Lévy, DE and
 265 baseline trial counts as one NFE; post-run diagnostics are excluded from and
 266 invisible to the search. Twenty pre-specified repetitions share a persisted
 267 initial population and its SHA-256 across all algorithms. The first member
 268 is the feasible M-A representation; the remaining members are generated by
 269 sampling the original 275-slot box and mapping it to quotient coordinates.
 270 Algorithm seeds are paired by repetition.

271 All confirmatory solvers minimize the same static target

$$F = 0.50838 C/C_{ref} + 0.49162 E/E_{ref} + 3P + 0.22L_{T1}/L. \quad (24)$$

272 Unlike the historical main search, no soft-to-hard penalty switch occurs; this
 273 prevents stale soft-stage fitness values from being compared with hard-stage
 274 values and makes feasibility handling identical across algorithms. The com-
 275 parison includes JS, GA, PSO, GWO and NSGA-II, while the IJS mech-
 276 anism study is the full 2^3 Tent–Lévy–DE factorial design. GA, PSO and

277 GWO retain the repository baseline equations and parameters, with only
 278 exact-NFE stopping added. NSGA-II searches the two penalized normalized
 279 objectives and selects the evaluated non-dominated candidate minimizing the
 280 same frozen weighted sum.

281 Primary paired contrasts use the two-sided Wilcoxon signed-rank test on
 282 F , with Holm correction within each pre-specified family. Reports also in-
 283 clude paired rank-biserial effects, median paired differences, 20,000-resample
 284 paired bootstrap 95% intervals, exact paired feasibility tests, raw samples and
 285 wall time. Secondary C and E contrasts are reported separately. Current-
 286 model structural ablations compare joint search with plan-only, profile-only
 287 and a 300,000+300,000-NFE sequential plan-profile search. Robustness re-
 288 runs use ten paired seeds for $W = 250/1000$ m, density-layer removal, raw
 289 versus road-reconstructed DEM, 50/100-m OSM bridge extensions and 20/40-
 290 m forced-structure thresholds. Operational fleet, demand and price scenarios
 291 are post-optimal evaluations, not re-optimizations.

292 5. Results

293 5.1. Current engineering scheme

294 Table 2 and Figure 4 report the current main result: $W = 500$ m, den-
 295 sity constraints active and endpoint elevations anchored. M-C reduces C
 296 by 8.33% and E by 3.00%. The route is 12 m longer (+0.05%), demon-
 297 strating that the saving is not an artificial consequence of a shorter path.
 298 Bridge/tunnel cost decreases by 237 million RMB, while earthwork rises by
 299 15 million RMB. The net effect is physically interpretable: the optimizer
 300 accepts more earthwork and a slightly longer route to reduce structural ex-
 301 posure and operating energy.

302 The reported diagnostic slope-hazard index is $Q = \sum_{u=1}^5 W_u S_u$ with
 303 scores $S_u \in \{1, 3, 5\}$ and weights (0.30, 0.15, 0.20, 0.20, 0.15) for topography,
 304 gully density, rainfall, geology and vegetation. The topographic score is the
 305 worse of grade-angle and cut/fill-height classes; the other four are corridor-
 306 level calibrated constants. Hence Q is a comparative screening diagnostic,
 307 not a probabilistic slope-failure model or an optimization objective.

308 The selected line crosses 1.591 km of Tier 1 and no Tier 2. Figure 5
 309 verifies this against the density field. It also avoids treating the existing route
 310 as an exception: Tier thresholds are calibrated so that a density already
 311 traversed by M-A is evidence of passability, while denser connected cores
 312 remain forbidden.

Table 2: Current endpoint-anchored, density-constrained result.

Metric	M-A existing	M-C optimized	Change
Life-cycle cost (10^8 RMB)	26.4061	24.2078	−8.33%
Traffic energy (10^8 RMB)	13.9460	13.5278	−3.00%
Fuel component	11.1072	10.7911	−2.85%
Electricity component	2.8388	2.7367	−3.60%
Length (km)	22.4618	22.4741	+0.05%
Minimum radius (m)	422	421	feasible
Slope-hazard index	3.309	3.298	−0.35%
Tier-1 exposure (km)	2.010	1.591	−20.85%
Tier-2 exposure (km)	0	0	feasible
Penalty	0	0	feasible

Cost and energy are 30-year discounted monetary quantities. Tier-1 is passable with a soft term; Tier-2 is forbidden.

313 The current Pareto set (Figure 6) contains costly structure-heavy energy
314 endpoints as well as the practically relevant cluster. Feasibility and the 10%
315 budget gate remove those extremes before entropy selection. The chosen
316 M-C is therefore a constrained compromise, not simply the minimum of a
317 globally normalized weighted sum.

318 Figure 7 provides the spatial interpretation. The present M-C contains
319 approximately 4.08 km of OSM-triggered crossing bridges and 0.75 km in
320 the Baiyun ecological tunnel zone. These values differ from earlier repository
321 figures because the current density-constrained solution is used here.

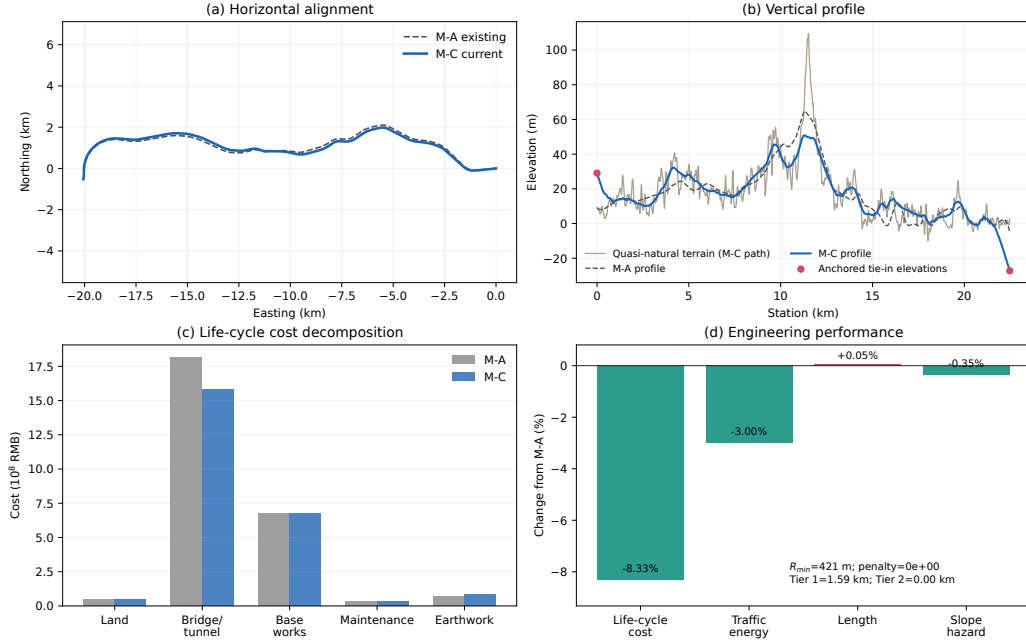


Figure 4: Current M-A/M-C comparison: (a) plan, (b) endpoint-anchored profile, (c) life-cycle cost decomposition and (d) headline changes. The increased earthwork and small length increase are shown explicitly.

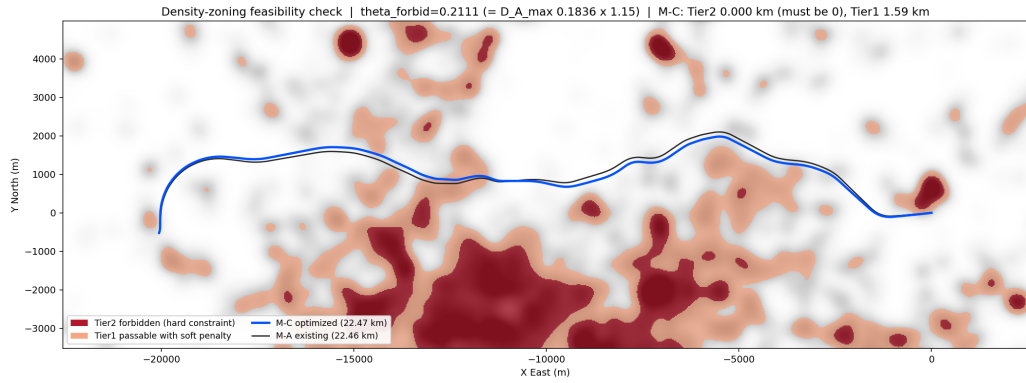


Figure 5: Building-density feasibility check. The current M-C line avoids Tier-2 cores and reduces Tier-1 exposure to 1.59 km.

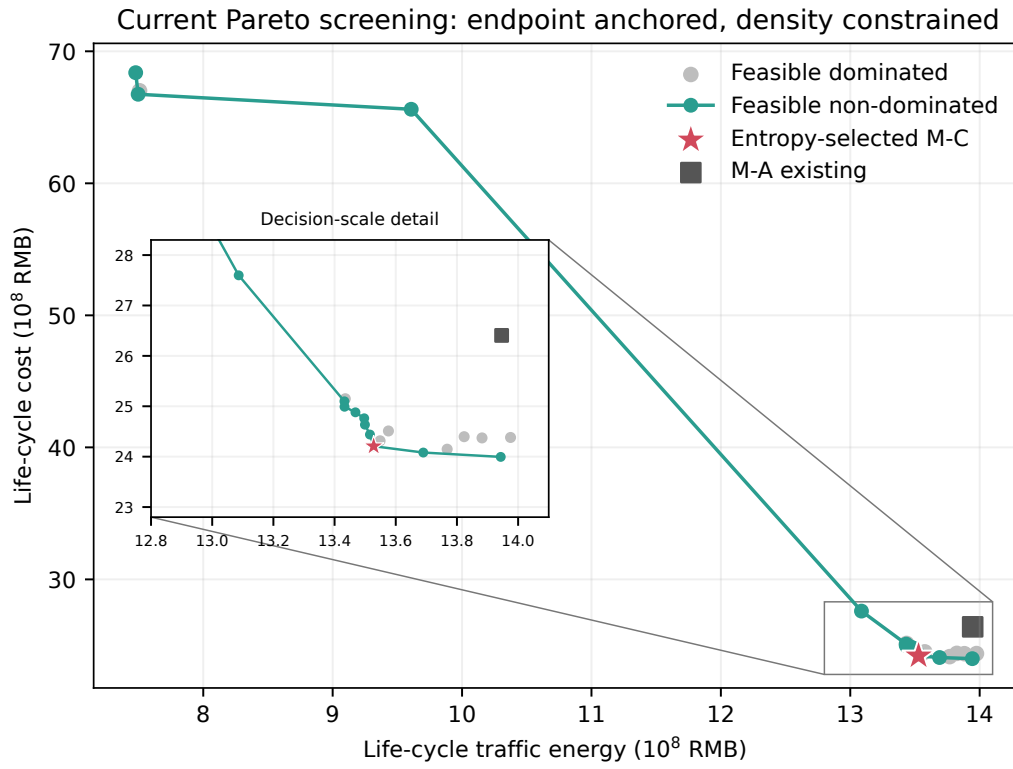


Figure 6: Feasible and non-dominated screening under the current endpoint and density constraints.

Current M-C alignment on quasi-natural DEM (endpoint anchored; density constrained)

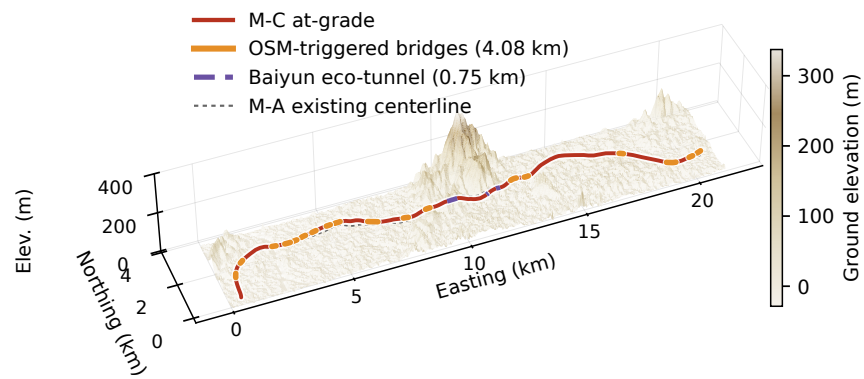


Figure 7: Current M-C on the quasi-natural DEM. At-grade, OSM-triggered bridge and ecological-tunnel intervals are distinguished; M-A is shown as a dashed reference.

322 5.2. Historical pre-density control: exploratory evidence

323 A controlled experiment completed before endpoint anchoring and density
 324 tiers used identical objective references and equal evaluation budgets for joint
 325 and two-stage methods. The existing line had $C = 26.4061$, $E = 13.9460$.
 326 The two-stage method obtained $C = 23.1582$ and $E = 13.4569$, apparently
 327 better than the joint result ($C = 24.2239$, $E = 13.4766$). However, its mini-
 328 mum radius was 397 m and penalty 7.36×10^{-2} , whereas the joint solution
 329 had 401 m and zero penalty. The lower two-stage objectives are therefore
 330 not admissible evidence of superiority. Freezing the plan after stage one
 331 removed degrees of freedom needed near the curvature boundary. This his-
 332 torical result motivates collaborative search, but must be rerun with current
 333 endpoint/density constraints before it can quantify a current-model effect.

334 5.3. Historical same-generation IJS ablation

335 Table 3 shows the 30-run historical ablation. At the same generation
 336 count, IJS lowers mean F from 0.9133 to 0.6808 and reaches the final JS level
 337 in 187 rather than 500 generations. The observed 25.45% gap is confounded
 338 with about three times the NFE and an earlier free-endpoint objective; it
 339 is not an equal-budget improvement estimate. DE has the largest observed
 340 isolated gap (24.17%) but also doubles NFE. Lévy and Tent show 3.79% and
 341 1.58% gaps. The unadjusted rank-sum p values are retained for audit, not
 342 confirmatory inference.

343 The corresponding observed-gap and run-distribution plots are retained
 344 as supplementary audit artifacts (`figures/ablation_contributions.pdf`
 345 and `ablation_boxplot.pdf`) rather than promoted as current-model confir-
 346 mation.

347 Mechanism instrumentation clarifies the historical observation. Indepen-
 348 dent Tent chains replace 132 of 200 random initial individuals. DE use is
 349 associated with about 0.018 lower F in the final 100 generations, consistent
 350 with a local-refinement hypothesis but not isolating it from the additional
 351 evaluations. A pre-defined “escape after 20 stagnant generations” statistic
 352 is not applicable because complete IJS rarely experiences a 20-generation
 353 stagnant interval; reporting it as zero would incorrectly imply failure.

354 5.4. Historical same-generation algorithm benchmark

355 At equal generation counts, IJS has the lowest historical mean F at all
 356 six scales (Table 4). These values describe an earlier free-endpoint model and
 357 IJS uses roughly three population-equivalent evaluations per generation, so

Table 3: Historical same-generation ablation. NFE differs by variant; p values are unadjusted independent-sample rank-sum tests.

Variant	Mean F	SD	F_{100}	Observed gap	p
JS	0.9133	0.0637	2.547	—	—
JS+Tent	0.8989	0.0605	2.227	1.58%	0.464
JS+Lévy	0.8787	0.0558	2.095	3.79%	0.0555
JS+DE	0.6926	0.0315	1.679	24.17%	3.02×10^{-11}
IJS	0.6808	0.0340	1.292	25.45%	3.02×10^{-11}

Table 4: Historical objective mean (SD) over ten runs. Equal generations do not mean equal NFE.

Algorithm	PJ1 (95)	PJ2	PJ3	PJ4	PJ5	PJ6 (500)
IJS	.6617 (.0051)	.5986 (.0045)	.7087 (.0045)	.7090 (.0033)	.7778 (.0078)	.8333 (.0019)
JS	.6771 (.0052)	.6104 (.0049)	.7237 (.0063)	.7215 (.0066)	.7954 (.0020)	.8563 (.0060)
NSGA-II	.6792 (.0045)	.6240 (.0057)	.7282 (.0053)	.7287 (.0080)	.8104 (.0084)	12.8859 (.7330)
GA	.7103 (.0088)	.6478 (.0171)	.7515 (.0090)	.7489 (.0071)	5.3906 (.5481)	25.6024 (1.0630)
PSO	.7023 (.0116)	.6370 (.0113)	.7436 (.0070)	.7632 (.0072)	9.1988 (.9109)	32.7437 (.6336)
GWO	.6843 (.0065)	.6194 (.0070)	.7233 (.0046)	.7276 (.0070)	1.1104 (.0158)	3.2157 (1.5706)

the table cannot establish equal-budget superiority. At PJ6, IJS and JS have 10/10 feasibility whereas the other four algorithms have 0/10; this feasibility pattern remains relevant as a hypothesis for a current-model, equal-NFE rerun.

The six-scale convergence and PJ6 distribution plots remain available as supplementary audit artifacts (`figures/algorithm_convergence.pdf` and `algorithm_pj6_boxplot.pdf`). They are omitted from the main narrative because unequal NFE prevents a confirmatory visual comparison.

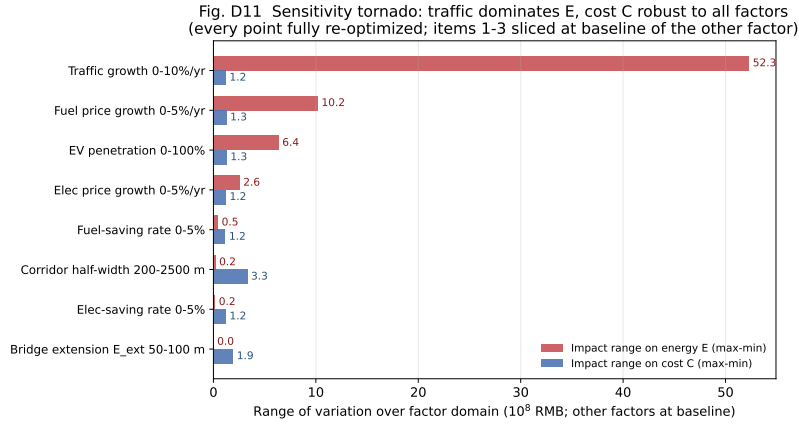
IJS takes about twice the wall time of one-phase competitors at PJ1–PJ5 because Lévy and DE add evaluations. The present record therefore supports only the statement that more evaluations per generation produced lower historical F . A current-model comparison must cap objective calls, pre-register paired seeds, report multiplicity-adjusted tests, effect sizes and confidence intervals, and include wall time.

5.5. Historical free-endpoint re-optimization sensitivity

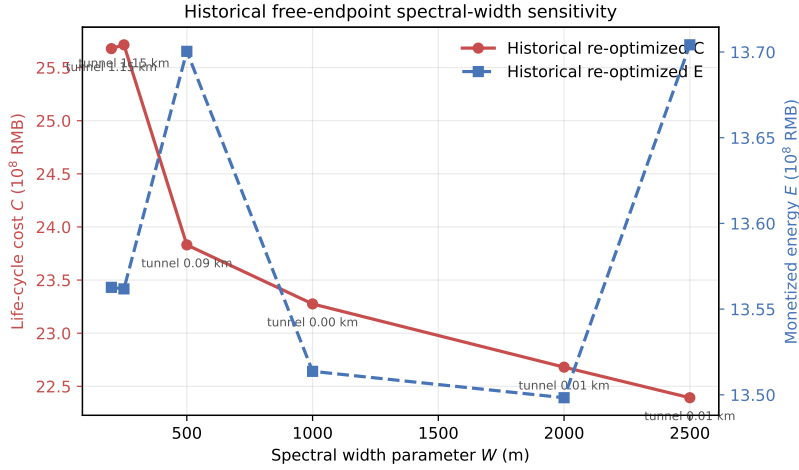
All 237 points in the historical free-endpoint model converge to feasible solutions. They do not validate robustness of the current endpoint/density formulation. Within that earlier model, traffic growth dominates the energy

376 response; cost is comparatively robust to demand, fleet and price factors
377 because infrastructure cost is incurred largely at construction. Raising EV
378 penetration moderates the high-growth energy burden, while simultaneous
379 fuel and electricity price growth increases the monetized energy objective.

380 The historical spectral-width parameter produces a strong then satu-
381 rating cost response: in the matched re-optimization experiment, C de-
382 creases from 2.568 billion RMB at $W = 200$ m to 2.239 billion RMB at
383 $W = 2,500$ m. The benefit accelerates once the spectral search can leave the
384 ecological-tunnel zone, then becomes limited by urban and geometric con-
385 straints. Varying bridge approach extension E_{ext} from 50/75/100 m shifts
386 C to 2.281/2.355/2.474 billion RMB, whereas E remains near 1.355 billion
387 RMB. Because this experiment uses the legacy free-endpoint model, it gen-
388 erates hypotheses rather than demonstrating current-model robustness.



(a) Relative importance of scenario factors.



(b) Historical spectral-width re-optimization; W is not a hard geometric band.

Figure 8: Historical free-endpoint sensitivity results. Every plotted point is re-optimized, but the experiment predates endpoint anchoring and density tiers.

389 6. Discussion

390 6.1. Engineering meaning of the improvement

391 The 8.33% cost decrease should not be interpreted as a generic saving
392 for all expressways. It is a case-specific comparison under common data and
393 prices. Its importance lies in the mechanism. M-C is slightly longer and
394 has higher earthwork than M-A, but it reduces structural cost and energy
395 while remaining outside Tier 2. The optimizer is therefore trading physical
396 components, not merely exploiting route length. The endpoint constraints
397 further prevent a candidate profile from receiving an unearned energy benefit
398 by starting high and ending low.

399 The pre-density two-stage control illustrates why objective values alone
400 are insufficient. Its lower C and E result from a line with $R_{min} = 397$ m.
401 Once feasibility is enforced, the claimed advantage disappears. Collaborative
402 search preserves the ability to relocate horizontal crossings while reshaping
403 the profile, which is particularly relevant where an OSM-triggered bridge and
404 a grade transition interact.

405 6.2. Role of DEM and OSM

406 DEM changes the objective landscape because the ground is sampled un-
407 der every candidate plan. Without it, lateral motion would affect only length
408 and abstract land price. OSM makes structural demand and urban exposure
409 geometry-dependent. The approach is not a substitute for survey-grade de-
410 sign: the dataset lacks bridge-deck elevations, parcel acquisition costs and
411 complete building attributes. Its value is to move early-stage optimization
412 from a one-dimensional centerline profile toward a reproducible spatial screen-
413 ing model.

414 The building tiers also clarify a policy distinction. Tier 2 is an exclusion
415 because its density exceeds what the existing expressway demonstrates as
416 traversable. Tier 1 is not forbidden; it carries a soft discouragement. This
417 avoids the unrealistic claim that an urban expressway can never pass de-
418 veloped land while still steering alternatives away from dense clusters. In
419 practice the soft term should be replaced by parcel-level acquisition and re-
420 settlement data.

421 6.3. Algorithmic interpretation

422 The historical ablation does not support a simplistic “chaos improves
423 everything” narrative: Tent has the smallest observed gap and DE the

largest. However, component activation also changes NFE, so the current evidence cannot separate operator quality from budget. Tent coverage, Lévy exploration and DE refinement remain plausible mechanisms to test under objective-call stopping.

The historical benchmark suggests that feasibility should be a primary algorithmic outcome. At high resolution several competitors return penalty-dominated solutions, but whether IJS retains that advantage at equal NFE under the current formulation is unresolved. Surrogate models, parallel evaluation or decomposition are promising routes to reduce cost after a fair baseline is established.

6.4. Limitations and validity threats

The following limitations are material rather than cosmetic.

- (1) The quasi-natural DEM is reconstructed by removing the present road influence, not observed before-construction terrain. Its 8.8-m pixel spacing cannot resolve local drainage and retaining structures.
- (2) OSM coverage is heterogeneous. Secondary roads do not trigger bridges, and bridge vertical clearance is not constrained because obstacle deck elevations are unavailable. A 15-degree skew limit is calibrated against the existing minimum of 15.6 degrees.
- (3) The Tier-1 weight (0.22 in normalized F) is a policy discouragement, not a measured demolition bill. Earlier ablation, benchmark and sensitivity experiments use both a pre-density and free-endpoint objective and unequal NFE; they are exploratory development records, not validation of the current scheme.
- (4) The energy objective is monetary and evaluates a representative travel direction. Endpoint anchoring removes net-elevation arbitrage, but a two-direction traffic-weighted model with calibrated regenerative braking is needed for final design.
- (5) Adjacent grade change is a proxy for vertical-curve smoothness. Explicit crest/sag curve lengths, non-overlap, sight distance, clothoid/arc segmentation and complete horizontal-vertical coordination checks are not yet implemented.

- 456 (6) Cost rates represent early-stage estimates. Geological investigation, hy-
 457 drology, parcel ownership, construction staging, embodied carbon and
 458 uncertainty are outside the current model.
- 459 (7) The selected current scheme is one optimization outcome. It requires
 460 multi-seed replication; algorithm claims require equal-NFE reruns, mul-
 461 tiplicity control, effect sizes and confidence intervals. The stored profile
 462 vector also retains one inactive legacy gene that should be removed in
 463 the next experiment release.

464 These limitations define the boundary of the claim: the framework sup-
 465 ports auditable single-run corridor screening and hypothesis generation; it
 466 does not yet support confirmatory method comparison or replace detailed
 467 geometric and construction design.

468 7. Conclusions

469 This study presents a data-driven, collaborative horizontal-vertical high-
 470 way alignment framework that connects three-dimensional geometry to life-
 471 cycle cost and mixed fuel-electric traffic energy. The model extends an ear-
 472 lier mathematical base with quasi-natural DEM sampling, OSM-triggered
 473 bridges, an ecological tunnel zone, building-density tiers, structural substi-
 474 tution and exact endpoint tie-ins. IJS supplies the implemented candidate
 475 search process, while feasibility and non-dominance precede entropy-based
 476 decision; its comparative advantage remains a confirmatory-experiment hy-
 477 pothesis.

478 For Guangzhou North Ring Expressway, the current $W = 500$ -m density-
 479 constrained result reduces cost by 8.33% and monetized energy by 3.00%,
 480 with $R_{min} = 421$ m, zero hard penalty and no Tier-2 penetration. The line is
 481 slightly longer and uses more earthwork, so the improvement is attributable
 482 to a transparent structural/operational trade rather than length alone. A
 483 two-stage historical control shows that superficially better objectives can be
 484 infeasible. Earlier ablation, benchmark and sensitivity experiments docu-
 485 ment the development path, but their free-endpoint model and unequal NFE
 486 prevent them from validating the current formulation. Equal-budget multi-
 487 seed replication is therefore a prerequisite for confirmatory algorithm claims.

488 The next research stage should implement bidirectional energy, survey-
 489 grade obstacle elevations, explicit vertical curves and sight distance,

490 parcel/embodied-carbon objectives, uncertainty-aware costs and multi-seed
491 replication of the latest density-constrained scheme. These additions would
492 move the framework from auditable corridor planning toward design-grade
493 decision support.

494 **Data and code availability**

495 The manuscript is generated from the RoadPlan experiment repository.
496 GNSS and project-derived files may be subject to owner restrictions; pro-
497 cessed DEM/OSM layers, source code, fixed random seeds, result JSON
498 files and figure-generation scripts are organized for reproducibility. The lo-
499 cal package includes `SUPPLEMENT_REPRODUCIBILITY.md`, a source-value snap-
500 shot and a SHA-256 manifest. A public archive and DOI should be inserted
501 after anonymization review: `[DATA-REPOSITORY-DOI-PLACEHOLDER]`.

502 **CRedit authorship contribution statement**

503 **[Author name placeholders]:** Conceptualization, Methodology,
504 Software, Validation, Investigation, Data curation, Visualization, Writing—
505 original draft, Writing—review & editing. Contributions must be replaced
506 with the actual author roles before submission.

507 **Declaration of competing interest**

508 The authors declare that they have no known competing financial inter-
509 ests or personal relationships that could have appeared to influence the work
510 reported in this paper. This statement must be confirmed by all authors
511 before submission.

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515 **References**

- 516 [1] E. P. Chew, C. J. Goh, T. F. Fwa, Simultaneous optimization of horizon-
517 tal and vertical alignments for highways, *Transportation Research Part*
518 *B: Methodological* 23 (5) (1989) 315–329. [doi:10.1016/0191-2615\(89\)](https://doi.org/10.1016/0191-2615(89)90008-8)
519 [90008-8](https://doi.org/10.1016/0191-2615(89)90008-8).

- 520 [2] J.-C. Jong, P. Schonfeld, An evolutionary model for simultaneously
521 optimizing three-dimensional highway alignments, *Transportation Re-*
522 *search Part B: Methodological* 37 (2) (2003) 107–128. doi:[10.1016/
523 S0191-2615\(01\)00047-9](https://doi.org/10.1016/S0191-2615(01)00047-9).
- 524 [3] M.-W. Kang, M. K. Jha, P. Schonfeld, Applicability of highway align-
525 ment optimization models, *Transportation Research Part C: Emerging*
526 *Technologies* 21 (1) (2012) 257–286. doi:[10.1016/j.trc.2011.09.006](https://doi.org/10.1016/j.trc.2011.09.006).
- 527 [4] S. Mondal, Y. Lucet, W. Hare, Optimizing horizontal alignment of roads
528 in a specified corridor, *Computers & Operations Research* 64 (2015) 130–
529 138. doi:[10.1016/j.cor.2015.05.018](https://doi.org/10.1016/j.cor.2015.05.018).
- 530 [5] D. Hirpa, W. Hare, Y. Lucet, Y. Pushak, S. Tesfamariam, A bi-objective
531 optimization framework for three-dimensional road alignment design,
532 *Transportation Research Part C: Emerging Technologies* 65 (2016) 61–
533 78. doi:[10.1016/j.trc.2016.01.016](https://doi.org/10.1016/j.trc.2016.01.016).
- 534 [6] M. E. Vázquez-Méndez, G. Casal, D. Santamarina, A. Castro, A 3d
535 model for optimizing infrastructure costs in road design, *Computer-*
536 *Aided Civil and Infrastructure Engineering* 33 (5) (2018) 423–439. doi:
537 [10.1111/mice.12350](https://doi.org/10.1111/mice.12350).
- 538 [7] A. Akhmet, W. Hare, Y. Lucet, Bi-objective optimization for road ver-
539 tical alignment design, *Computers & Operations Research* 143 (2022)
540 105764. doi:[10.1016/j.cor.2022.105764](https://doi.org/10.1016/j.cor.2022.105764).
- 541 [8] C. Han, T. Han, T. Ma, Z. Tong, S. Wang, T. Hei, End-to-end bim-based
542 optimization for dual-objective road alignment design with driving safety
543 and construction cost efficiency, *Automation in Construction* 151 (2023)
544 104884. doi:[10.1016/j.autcon.2023.104884](https://doi.org/10.1016/j.autcon.2023.104884).
- 545 [9] Q. He, T. Gao, Y. Gao, H. Li, P. Schonfeld, Y. Zhu, Q. Li, P. Wang, A bi-
546 objective deep reinforcement learning approach for low-carbon-emission
547 high-speed railway alignment design, *Transportation Research Part C:*
548 *Emerging Technologies* 147 (2023) 104006. doi:[10.1016/j.trc.2022.
549 104006](https://doi.org/10.1016/j.trc.2022.104006).
- 550 [10] B. M. A. Al-Hadad, W. H. Nadir, G. A. M. Jukil, Intelligent optimiza-
551 tion of highway alignments: A novel approach integrating geographic

- information system and genetic algorithms, *Engineering Applications of Artificial Intelligence* 133 (2024) 108037. [doi:10.1016/j.engappai.2024.108037](https://doi.org/10.1016/j.engappai.2024.108037).
- [11] H. Pu, X. Wan, T. Song, P. Schonfeld, L. Peng, A 3d-rrt-star algorithm for optimizing constrained mountain railway alignments, *Engineering Applications of Artificial Intelligence* 130 (2024) 107770. [doi:10.1016/j.engappai.2023.107770](https://doi.org/10.1016/j.engappai.2023.107770).
- [12] H. Pu, Q. Zeng, T. Song, P. Schonfeld, G. Wang, W. Li, L. Peng, J. Hu, A hybrid proximal policy optimization and particle swarm algorithm for highway alignment optimization, *Advanced Engineering Informatics* 69 (2026) 103959. [doi:10.1016/j.aei.2025.103959](https://doi.org/10.1016/j.aei.2025.103959).
- [13] T. Song, P. Schonfeld, H. Pu, A review of alignment optimization research for roads, railways and rail transit lines, *IEEE Transactions on Intelligent Transportation Systems* 24 (5) (2023). [doi:10.1109/TITS.2023.3235685](https://doi.org/10.1109/TITS.2023.3235685).
- [14] K. Lin, A three-dimensional alignment design method for highways with life-cycle cost and energy consumption objectives, Master's thesis, Foshan University, Foshan, China (2025).
- [15] N. Davey, S. Dunstall, S. Halgamuge, Optimal road design through ecologically sensitive areas considering animal migration dynamics, *Transportation Research Part C: Emerging Technologies* 77 (2017) 478–494. [doi:10.1016/j.trc.2017.02.016](https://doi.org/10.1016/j.trc.2017.02.016).
- [16] H. Zhang, H. Pu, P. Schonfeld, T. Song, W. Li, J. Wang, X. Peng, J. Hu, Multi-objective railway alignment optimization considering costs and environmental impacts, *Applied Soft Computing* 89 (2020) 106105. [doi:10.1016/j.asoc.2020.106105](https://doi.org/10.1016/j.asoc.2020.106105).
- [17] T. Song, H. Pu, P. Schonfeld, H. Zhang, W. Li, J. Hu, Simultaneous optimization of 3d alignments and station locations for dedicated high-speed railways, *Computer-Aided Civil and Infrastructure Engineering* 37 (4) (2022) 405–426. [doi:10.1111/mice.12739](https://doi.org/10.1111/mice.12739).
- [18] D. Llopis-Castelló, F. J. Camacho-Torregrosa, A. García, Analysis of the influence of geometric design consistency on vehicle co2 emissions,

- 584 Transportation Research Part D: Transport and Environment 69 (2019)
585 40–50. [doi:10.1016/j.trd.2019.01.029](https://doi.org/10.1016/j.trd.2019.01.029).
- 586 [19] A. Di Martino, S. M. Miraftebzadeh, M. Longo, Strategies for the mod-
587 elisation of electric vehicle energy consumption: A review, *Energies*
588 15 (21) (2022) 8115. [doi:10.3390/en15218115](https://doi.org/10.3390/en15218115).
- 589 [20] K. Deb, A. Pratap, S. Agarwal, T. Meyarivan, A fast and elitist multiob-
590 jective genetic algorithm: Nsga-ii, *IEEE Transactions on Evolutionary*
591 *Computation* 6 (2) (2002) 182–197. [doi:10.1109/4235.996017](https://doi.org/10.1109/4235.996017).
- 592 [21] J.-S. Chou, D.-N. Truong, A novel metaheuristic optimizer inspired by
593 behavior of jellyfish in ocean, *Applied Mathematics and Computation*
594 389 (2021) 125535. [doi:10.1016/j.amc.2020.125535](https://doi.org/10.1016/j.amc.2020.125535).
- 595 [22] R. Storn, K. Price, Differential evolution—a simple and efficient heuris-
596 tic for global optimization over continuous spaces, *Journal of Global*
597 *Optimization* 11 (1997) 341–359. [doi:10.1023/A:1008202821328](https://doi.org/10.1023/A:1008202821328).
- 598 [23] C. E. Shannon, A mathematical theory of communication, *Bell System*
599 *Technical Journal* 27 (3) (1948) 379–423. [doi:10.1002/j.1538-7305.](https://doi.org/10.1002/j.1538-7305.1948.tb01338.x)
600 [1948.tb01338.x](https://doi.org/10.1002/j.1538-7305.1948.tb01338.x).
- 601 [24] Amazon Web Services, [Terrain tiles on the registry of open data on aws](https://registry.opendata.aws/terrain-tiles/),
602 AWS Open Data Registry (2026).
603 URL <https://registry.opendata.aws/terrain-tiles/>
- 604 [25] M. Haklay, P. Weber, Openstreetmap: User-generated street maps,
605 *IEEE Pervasive Computing* 7 (4) (2008) 12–18. [doi:10.1109/MPRV.](https://doi.org/10.1109/MPRV.2008.80)
606 [2008.80](https://doi.org/10.1109/MPRV.2008.80).
- 607 [26] OpenStreetMap Foundation, [Openstreetmap copyright and license](https://www.openstreetmap.org/copyright)
608 (2026).
609 URL <https://www.openstreetmap.org/copyright>
- 610 [27] Guangzhou Municipal Development and Reform Commission, Guide-
611 lines for estimating municipal government-invested projects: Volume ii,
612 municipal transportation engineering, Tech. rep., Guangzhou Muni-
613 cipal Government, Guangzhou, official Chinese cost-estimation guideline;
614 archived with the experiment data (2025).

- 615 [28] Ministry of Transport of the People's Republic of China, Design Specifi-
616 cation for Highway Alignment (JTG D20-2017), China Communications
617 Press, Beijing (2017).